Designing a Minimal Data-Driven Dashboard for Software Project Health and Delivery Control

ITCS514 Software Project Management (1/2025)

Emerging Frontiers in Software Project Management Individual Assignment

Thant Zin Myo

Contents

1. Introduction 3
1.1 Background and Motivation 3
1.2 Research Questions 4
1.3 Research Objectives 4
1.4 Significance of the Study 4
2. Background and Literature Review 5
2.1 Software Project Failure: Causes and Consequences 5
2.2 Project Management Metrics and KPIs 5
2.3 Data-Driven Decision Making in SMEs 5
2.4 Research Gap 6
3. Methodology and Analytical Framework 6
3.1 Research Approach 6
3.2 Case Study Context 6
3.3 Data Collection 7
3.4 Analytical Framework 7
4. Findings and Discussion 7
4.1 Essential Metric Categories 7
4.1.1 Schedule Performance Metrics 8
4.1.2 Budget and Cost Management Metrics 9
4.1.3 Quality Assurance Metrics 10
4.1.4 Risk Management Metrics 11
4.1.5 Stakeholder Engagement Metrics 11
4.1.6 Change Management Metrics 12
4.2 Metric Interdependencies and Relationships 13
4.3 Practical Implementation Insights 13
5. Conclusion and Recommendations 13
5.1 Summary of Findings 13
5.2 Recommendations 14
5.3 Limitations 14
5.4 Final Thoughts 14

References 15

Table of Figures

Figure 1: The six essential metric categories forming the minimal dashboard framework. 8
Figure 2: Timeline Dashboard to keep track of activities and deliverables 8
Figure 3: Budget estimation dashboard with two different approaches 9
Figure 4: Budget spending analysis dashboard 9
Figure 5: Quality Score 10
Figure 6: Quality Weight Calculation 10
Figure 7: Risk analysis Dashboard 11
Figure 8: Stakeholder Management Dashboard 12
Figure 9: Change Request Dashboard 12

1. Introduction

1.1 Background and Motivation

The software development industry faces a persistent challenge: project failure rates remain alarmingly high despite decades of methodological improvements and technological advancements. According to the (Group, 2020)Standish Group's CHAOS Report, only 29% of software projects are considered successful, while 52% are challenged (over budget, behind schedule, or with reduced functionality), and 19% fail completely (The Standish Group, (Group, 2020)). These statistics represent not only financial losses but also missed opportunities, damaged stakeholder relationships, and organizational inefficiencies.

The main root causes of project failures are multifaceted, ranging from poor requirements management and inadequate stakeholder communication to insufficient risk monitoring and budget overruns. However, a common thread emerges from failure analyses: the lack of timely, actionable information about project health. Many organizations either lack systematic data collection mechanisms or are overwhelmed by excessive metrics that obscure rather than illuminate critical issues (Verner, J. M.; Brereton, O. P.; Kitchenham, B. A.; Turner, M.; Niazi, M., 2014).

While large enterprises can afford sophisticated project management information systems (PMIS), startups and small-to-medium enterprises (SMEs) often operate with limited resources, tools, and planning capacity (Tawil, Bach, Roushan, & Al-Karaghouli, 2024) (Parizotto, Tonso, & Carvalho, 2020). These organizations need lightweight, practical solutions that provide maximum insight with minimal overhead. This creates a compelling need for a "minimal" data-driven approach which focuses on essential metrics that truly matter for project success.

1.2 Research Questions

This research addresses three key questions: what minimal yet essential metrics are required to effectively monitor software project health and delivery performance, how these metrics can be organized and visualized in a dashboard format that supports rapid decision-making and early risk detection, and what practical insights can be derived from implementing such a minimal dashboard in a real software project context.

1.3 Research Objectives

The primary objectives of this study are to identify and justify a core set of software project metrics through both literature review and empirical analysis, to design a minimal data-driven dashboard framework that balances comprehensiveness with simplicity, to demonstrate the practical application of this framework through a case study of a campus event management system project, and to provide actionable recommendations for software project managers, particularly those working in resource-constrained environments.

1.4 Significance of the Study

This research contributes to both academic knowledge and professional practice in several ways. First, it synthesizes existing research on software project metrics into a practical, minimal framework suitable for SMEs and startups. Second, it provides empirical evidence from a real project implementation, demonstrating how theoretical concepts translate into practice.

2. Background and Literature Review

2.1 Software Project Failure: Causes and Consequences

Research by (Verner, J. M.; Brereton, O. P.; Kitchenham, B. A.; Turner, M.; Niazi, M., 2014) identified several primary causes of software project failure through a comprehensive survey of project managers. The top factors included poor requirements management, inadequate stakeholder engagement, unrealistic schedules and budgets, insufficient risk management, quality issues, and scope creep. Importantly, (Verner, J. M.; Brereton, O. P.; Kitchenham, B. A.; Turner, M.; Niazi, M., 2014) noted that many of these issues could be detected and addressed early if proper monitoring mechanisms were in place. This finding underscores the importance of systematic project health monitoring.

2.2 Project Management Metrics and KPIs

The concept of measuring project performance through metrics and key performance indicators (KPIs) is well-established in project management literature. The Project Management Institute's PMBOK Guide (Project Management Institute, 2021) emphasizes the importance of monitoring and controlling processes, which rely heavily on performance data.

(Kerzner, 2017) categorizes project metrics into several dimensions: schedule metrics, cost metrics, quality metrics, resource metrics, and risk metrics. However, (Kerzner, 2017) also warns against "metric overload," where organizations collect excessive data that becomes burdensome to maintain and difficult to interpret. This observation is particularly relevant for SMEs with limited resources.

2.3 Data-Driven Decision Making in SMEs

Recent research (Tawil, Bach, Roushan, & Al-Karaghouli, 2024) examined trends and challenges in data-driven decision-making among UK SMEs. Their analysis of 85 organizations revealed that SMEs recognize the value of data-driven approaches but face significant implementation barriers including resource constraints, limited technical capabilities, and cultural factors. These findings reinforce the need for minimal, practical approaches to project monitoring that can be implemented without extensive infrastructure or specialized expertise (Abdullahi, A.; Sarker, S., 2023).

2.4 Research Gap

While extensive literature exists on project metrics, and dashboard design, there is limited research specifically addressing the question of "minimal sufficiency" in project monitoring. Most frameworks assume either large-scale enterprise contexts or focus on specific methodologies (Agile, Waterfall) rather than providing a methodology-agnostic, minimal approach suitable for resource-constrained organizations. This research addresses this gap by synthesizing existing knowledge into a practical, minimal framework and demonstrating its application through empirical case study analysis.

3. Methodology and Analytical Framework

3.1 Research Approach

This study employs a mixed-methods approach combining literature review, framework development, case study analysis, and comparative analysis. The literature review systematically analyzes academic research and industry best practices related to software project metrics and dashboard design. Framework development synthesizes literature findings into a minimal dashboard framework identifying essential metric categories. The case study provides empirical examination of a real software project to validate and refine the framework.

3.2 Case Study Context

The empirical component of this research analyzes a campus event management system project with the following characteristics:

- Project Type: Web-based application development
- Duration: Approximately 3.5 months (August 2025 - November 2025)
- Team Size: Small team including project manager, system architect, developers, and QA testers
- Budget: $500,00 (including labor and overhead costs)
- Methodology: Hybrid approach incorporating elements of traditional project management

This project provides a realistic context for examining how minimal dashboard metrics can be applied in practice.

3.3 Data Collection

Data was collected from eight primary project management documents: Project Selection, Stakeholder Management, Project Schedule, Project Budget, Project Risk, Change Request, Quality Metrics, and Project Overview excel sheets. These excel sheet captured comprehensive data across all dimensions of project management, providing rich material for analysis.

3.4 Analytical Framework

The analysis follows a structured four-step approach. First, Metric Identification involves extracting all metrics tracked across the Excel files and categorizing them by dimension. Second, Essentiality Assessment evaluates each metric category based on its predictive power and actionability. Third, Dashboard Design organizes the essential metrics into a hierarchical dashboard structure with appropriate visualizations. Finally, Validation uses actual project data to assess the dashboard's effectiveness and identify insights that could support earlier intervention.

4. Findings and Discussion

4.1 Essential Metric Categories

Through both literature review and empirical analysis, this research identifies six essential metric categories required for a minimal software project dashboard: Budget Utilization, Schedule Progress, Scope Completion, Quality Score, Risk Score, and Overall Performance. These categories represent the minimum information needed to effectively monitor project health and delivery control. Budget, Schedule, and Risk each have their own dedicated dashboards due to their importance and the level of detail they require. Scope Completion can be tracked through the schedule dashboard and supplemented by a configuration and change management dashboard where necessary. A stakeholder management dashboard is also included, supported by relevant literature, given its close relationship to the six core categories and its influence on project outcomes. Finally, an overall dashboard provides a consolidated view of all key metrics, enabling rapid assessment of project status at a glance.

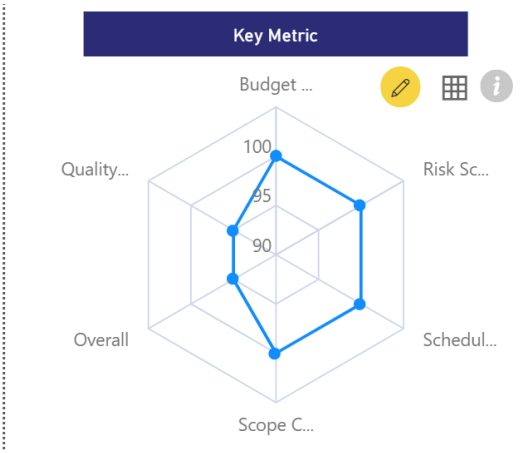


Figure 1: The six essential metric categories forming the minimal dashboard framework.

4.1.1 Schedule Performance Metrics

Schedule performance emerged as the most fundamental metric category, as time is often the most visible and consequential constraint in software projects. The case study project tracked 10 major activities with planned vs. actual dates, progress percentages, and variance analysis.

Key Findings: The project experienced 2 delayed activities (Requirements Gathering and System Analysis), each delayed by 1-2 days. Despite early delays, the project maintained and managed to closure the project 2 days early.

Essential Schedule Metrics include total activities vs. completed activities, number of delayed activities, schedule variance (planned vs. actual), and overall schedule health indicator.

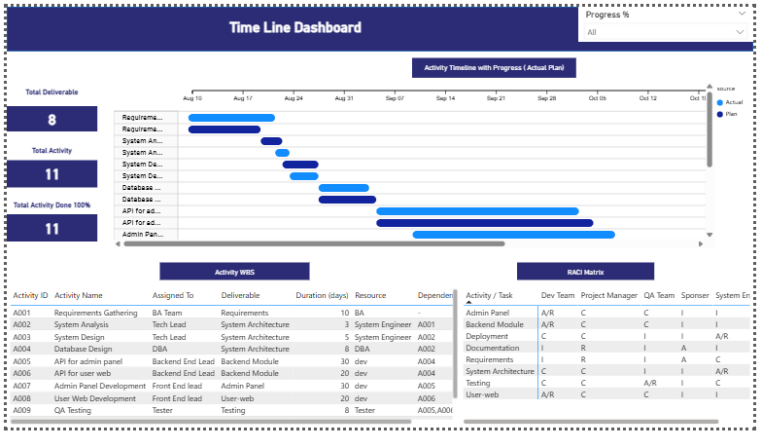


Figure 2: Timeline Dashboard to keep track of activities and deliverables

4.1.2 Budget and Cost Management Metrics

Financial performance tracking is critical for project viability and organizational sustainability. The case study employed two estimation methods (resource-based and function point analysis) and tracked actual costs against estimates.

Insights: The case study revealed that using multiple estimation methods provided validation and increased confidence in budget projections. Resource-based estimated around $49,537 and Function point estimate around 47974. The small variance (4%) between methods suggested good estimation practices. So make it simpler and backup risk cover budget Project Manager choose to estimate with $50,000. Even thought The budget estimate around $50,000, Actual spending was only around $41900 only because we make profit out of admin overhead and contingency since no much risk and external unknow charges occurs . Essential Budget Metrics include total budget vs. actual spending, budget utilization percentage, cost variance by category, burn rate, and budget health indicator.

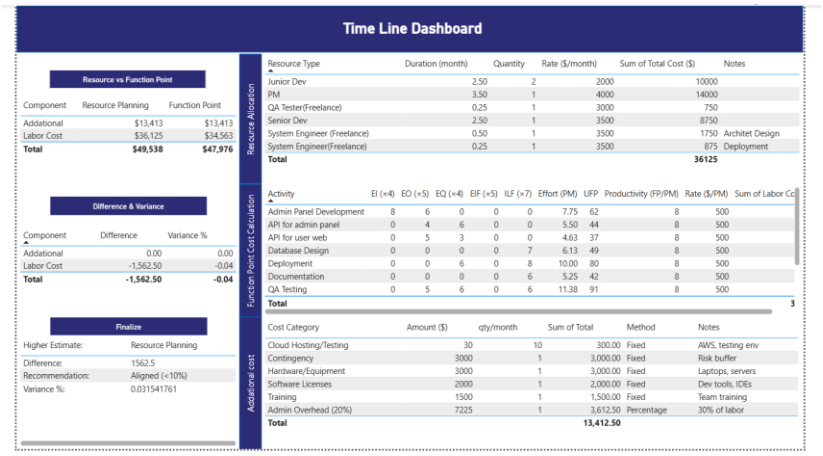


Figure 3: Budget estimation dashboard with two different approaches

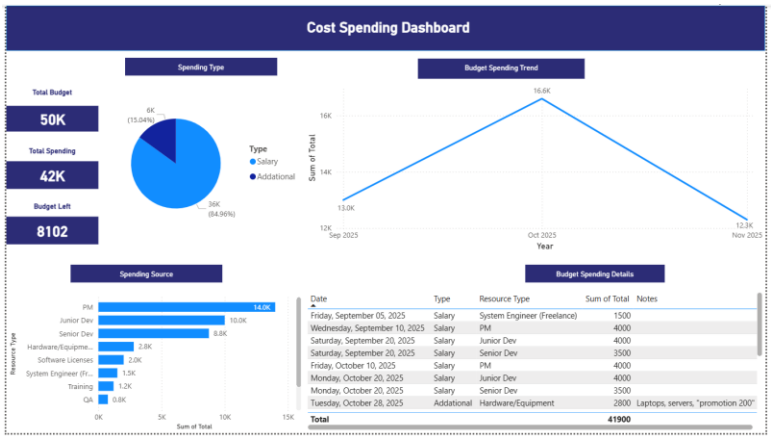


Figure 4: Budget spending analysis dashboard

4.1.3 Quality Assurance Metrics

Quality metrics assess whether the project delivers value that meets stakeholder expectations. The case study tracked quality across ISO25010 quality metric dimensions (ISO25010, 2011): Functional Suitability, Performance Efficiency, Compatibility, Interaction Compability, Reliability, and Security.

Insight: The weighted scoring approach allowed the project to prioritize different quality dimensions based on stakeholder importance. For example, Performance Efficiency received a 15% weight while Compatibility received 10%, reflecting the relative importance of these characteristics for the campus event application.

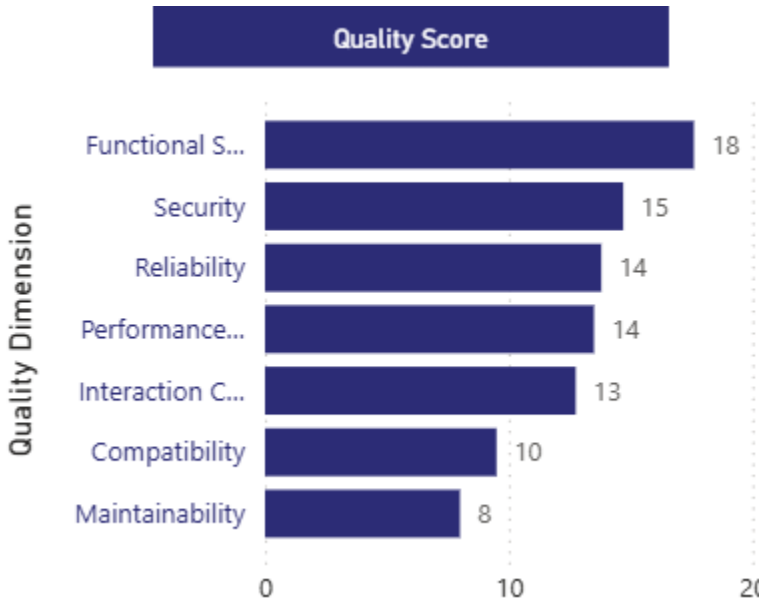


Figure 5: Quality Score

Quality Metrics

Quality Dimension	Targe	Curren	Status	Weigh	Weighted Scor
Functional Suitability	90%	88%	Good	20%	17.60
Performance Efficiency	85%	90%	Excellent	15%	13.50
Compatibility	95%	95%	Excellent	10%	9.50
Interaction Capability	90%	85%	Good	15%	12.75
Reliability	95%	92%	Good	15%	13.80
Security	100%	98%	Excellent	15%	14.70
Maintainability	85%	80%	Acceptable	10%	8.00

Figure 6: Quality Weight Calculation

4.1.4 Risk Management Metrics

Proactive risk management distinguishes successful projects from failed ones. The case study identified and tracked 3 major risks technical, resource categories. Essential Risk Metrics include total risks identified, high-priority risk count, open vs. solved risks, risk score distribution, and risk health indicator.

Insight: The case study revealed that 100% of identified risks were solved



Figure 7: Risk analysis Dashboard

4.1.5 Stakeholder Engagement Metrics

Stakeholder management is essential because it ensures that the expectations, concerns, and priorities of key individuals are continuously aligned with project goals. Effective engagement reduces risks, improves decision-making, and supports smoother delivery by keeping all critical contributors informed and involved.

The project engaged five key stakeholders with different power and interest levels, resulting in 24 meetings from August to December 2025. High power stakeholders such as the Project Sponsor, the System Engineer, and the Dev Team Lead were managed closely through weekly or biweekly syncs focused on progress, infrastructure readiness, and integration. The QA Lead was kept informed through weekly QA reviews, while the Student Union was monitored through monthly sessions centered on feature validation and user feedback.

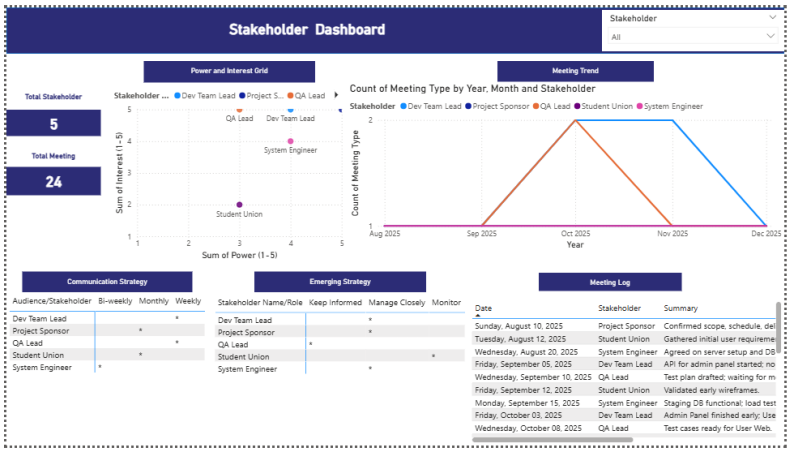


Figure 8: Stakeholder Management Dashboard

4.1.6 Change Management Metrics

Uncontrolled scope changes are a leading cause of project failure. The case study tracked 2 formal change requests with comprehensive impact analysis.

Insight: The case study demonstrated effective change control with formal impact analysis for each request. For example, the database technology change (MySQL to PostgreSQL) was evaluated for technical impact, benefits (JSON query support), and effort (same as original estimate), enabling informed decision-making.

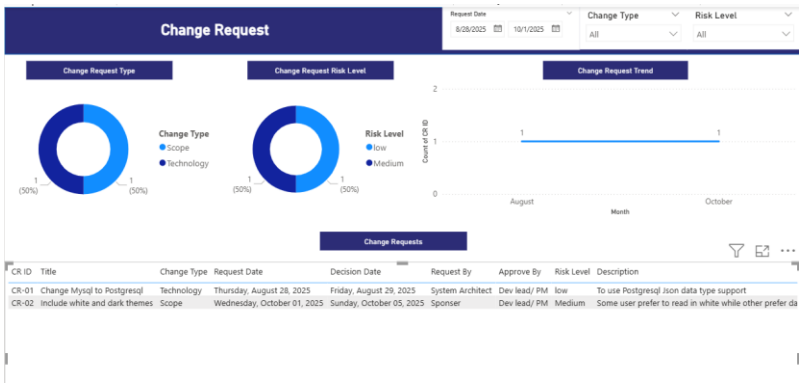


Figure 9: Change Request Dashboard

4.2 Metric Interdependencies and Relationships

Analysis of the case study data revealed important interdependencies among metric categories. Schedule delays often correlate with budget overruns due to extended resource allocation. Technical risks directly impact quality metrics. Change requests create ripple effects across schedule and budget. Stakeholder engagement influences quality through requirements clarity and feedback. These interdependencies suggest that the six metric categories should not be viewed in isolation but as an integrated system. Dashboard design should facilitate recognition of these relationships through cross-category views and correlation analysis.

4.3 Practical Implementation Insights

The case study implementation revealed several practical insights. The minimal dashboard approach required approximately 2-3 hours per week for data collection and updates across all six categories, representing a manageable overhead for small teams. The case study used Excel and BI tools for all tracking, demonstrating that sophisticated software is not required for effective dashboard implementation.

5. Conclusion and Recommendations

5.1 Summary of Findings

This research addressed the critical question of what constitutes a minimal yet effective set of metrics for monitoring software project health and delivery control. Through comprehensive literature review and empirical case study analysis, the study identified six essential metric categories: Schedule Performance, Budget and Cost Management, Quality Assurance, Risk Management, Stakeholder Engagement, and Change Management.

The case study of a campus event management system project demonstrated that these six categories provide adequate visibility into project health while maintaining manageable data collection overhead. Key findings include: the minimal dashboard approach requires approximately 2-3 hours per week for data maintenance; early warning signals were successfully detected across multiple metric categories; metric interdependencies exist and should be considered in dashboard design; the framework can be implemented with readily available tools.

5.2 Recommendations

Based on the research findings, several recommendations can be made. For organizations adopting minimal dashboards, it is advisable to begin with the essential categories rather than attempting broad, exhaustive measurement. Organizations should also customize their dashboards thoughtfully, tailoring specific metrics to their unique context, while integrating tools that consolidate data from multiple sources for greater efficiency. Establishing regular cadences for updates and review sessions is essential, along with ensuring that stakeholders are trained to correctly interpret and use the dashboard. The dashboard should be treated as a living system that is continuously improved over time. For project managers, recommendations include prioritizing actionability when selecting metrics, recognizing the interdependencies that exist across categories, and providing appropriate context when presenting dashboard data. It is also important to maintain a balance between leading and lagging indicators within each category to ensure well-rounded project visibility.

5.3 Limitations

This research has several limitations that should be acknowledged. The empirical component relies on a single project case study, which limits generalizability. The analysis was conducted retrospectively using existing project data. The case study represents an educational/institutional context, and different industries may have unique requirements. The project used Excel-based tools, which may have influenced data collection practices. The project duration (3.5 months) is relatively short, and longer projects might reveal different patterns.

5.4 Final Thoughts

Software project failure remains a persistent challenge, but it is not inevitable. This research demonstrates that effective project monitoring does not require sophisticated infrastructure or comprehensive measurement systems. By focusing on six essential metric categories, organizations can gain adequate visibility into project health while maintaining manageable overhead.

The minimal dashboard framework represents a practical, accessible approach to data-driven project management that is particularly well-suited for SMEs, startups, and resource-constrained organizations. However, its value extends beyond these contexts. Even large organizations with sophisticated PMIS systems can benefit from the clarity and focus that a minimal approach provides.

Ultimately, the goal of project monitoring is not to collect data for its own sake but to enable better decisions that increase the likelihood of project success. The minimal dashboard framework serves this goal by providing essential information in an accessible, actionable format. The path to reducing software project failure rates lies not in measuring everything but in measuring what matters.

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All supporting materials used in the framework including the Excel files, Power BI dashboards, and other analysis artifacts are made available for download from this link, ensuring full transparency and enabling replication or adaptation of the dashboard framework in future projects.